

Junzhao Yang

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EDUCATION

Carnegie Mellon University , Computer Science Department Ph.D. in Computer Science (expected 2027). Advisor: Richard Peng Research areas: numerical linear algebra, spectral graph theory	2024–Present
Tsinghua University , Institute for Interdisciplinary Information Sciences B.S. in Computer Science, Yao Class.	2020–2024

PROJECTS AND EXPERIENCE

FizzyProver Sole developer · Multi-agent system for mathematical research	May 2026–Present
- Designed a recursive Codex workflow that schedules independent proof subproblems in parallel and alternates prover and verifier passes to critique and revise proof attempts. - Built and deployed a web platform used by 20+ academic users across 300+ runs. Among the runs we have examined to date, including runs powered by GPT-5.5, we have not observed any incorrect final proofs; by contrast, our direct GPT-5.5 Pro attempts produced flawed proofs.	
Megvii Internship Research Intern	Winter 2021
- Reproduced a graph-neural-network post-training algorithm from a research paper in PyTorch. - Optimized its large-matrix computations for GPU memory efficiency; portions of the implementation were later reused by the team.	

HONORS AND AWARDS

- ICPC North America Championship 2025, 2nd place, Silver Medal.
- 46th ICPC World Finals, 10th place, Bronze Medal** (held in 2024).
- International Olympiad in Informatics 2019, 5th place, Gold Medal.**
- National Olympiad in Informatics 2018, Gold Medal.
- Yao Award, Bronze Medal (awarded to five Yao Class students in 2023).

PUBLICATIONS

Authors are listed alphabetically, as is customary in theoretical computer science.

[1] Fast Convergence of Gradient Descent on Random ℓ_p-Norm Regression Yang Liu, Richard Peng, Weina Wang, Junzhao Yang	FOCS 2026 (to appear)
[2] Approximate Spanning Tree Counting from Uncorrelated Edge Sets Yang P. Liu, Richard Peng, Junzhao Yang	FOCS 2026 (to appear)
[3] Entrywise Approximate Solutions for SDDM Systems in Almost-Linear Time Angelo Farfan, Mehrdad Ghadiri, Junzhao Yang	STOC 2026
[4] Entrywise Approximation for Matrix Inversion and Linear Systems Mehrdad Ghadiri, Hoai-An Nguyen, Junzhao Yang	SODA 2026
[5] Numerical Linear Algebra in Linear Space Yiping Liu, Hoai-An Nguyen, Junzhao Yang	SODA 2026
[6] The Cost of Parallelizing Boosting Xin Lyu, Hongxun Wu, Junzhao Yang	SODA 2024
[7] Tight Time-Space Lower Bounds for Constant-Pass Learning Xin Lyu, Avishay Tal, Hongxun Wu, Junzhao Yang	FOCS 2023